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Chat with Llama-2 (7B) from HuggingFace (Llama-2-7b-chat-hf)



Nimrita Koul · Follow

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Image Source: <https://www.deviantart.com/nilesclover/art/Llama-Inside-A-Popcorn-Container-Raised-Arms-1012494804>

LLama 2 is a family of pretrained and fine-tuned text generation models based on autoregressive, transformer architecture. Llama 2 comes with 3 models with 7 billion, 13 billion and 70 billion parameters. Llama 2 is an

opensource model which is at par in performance with ChatGPT. You can use Llama2 for commercial as well as research use in English.

Llama 2 was pretrained on 2 trillion tokens of data from publicly available sources. The fine-tuning data includes publicly available instruction datasets, as well as over one million new human-annotated examples.

In this article, I will demonstrate how to get started using [Llama-2-7b-chat 7 billion parameter Llama 2 which is hosted at HuggingFace](#) and is finetuned for helpful and safe dialog using supervised fine-tuning (SFT) and RLHF. The code is in Python. I have used Google Colab Free Tier T4 GPU for executing this code. It won't work on CPU. The model is around 14GB, so you may run out of CUDA memory on Colab free tier. However, with parameters of `model.generate()` as in this notebook, you will be successfully able to run the code.

To use models from HuggingFace you should have a HuggingFace account, and a valid access token (which is free of cost). Once you create your HuggingFace account you can create a new access token or use the one generated by default for you. You can [find your access token here](#).

Step 1: Install required libraries

```
!pip install accelerate\  
          protobuf\  
          sentencepiece\  
          torch\  
          git+https://github.com/huggingface/transformers\  
          huggingface_hub
```

Step 2: Import required libraries

```
import torch  
from transformers import AutoModelForCausalLM, AutoTokenizer, TextIteratorStreamer
```

Step 3: Login to your HuggingFace Account from Colab using huggingface_hub login

Run the code below in code cell of your Colab notebook, it will give you a login prompt as shown below, paste the access token you copied from your HF account and click on Login button.

```
from huggingface_hub import login  
login()
```

```
from huggingface_hub import notebook_login
```

```
notebook_login()
```

✓ 0.9s

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Copy a token from [your Hugging Face tokens page](#) and paste it below.

Immediately click login after copying your token or it might be stored in plain text in this notebook file.

Token:

Login

Step 4: Load the llama-2-7b-chat-hf model and the corresponding tokenizer. The llama2 models won't work on CPU so you must use GPU.

```
if torch.cuda.is_available():
    model_id = "meta-llama/Llama-2-7b-chat-hf"
    model = AutoModelForCausalLM.from_pretrained(model_id, torch_dtype=torch.float16, device_map="auto")
    model.cuda()
    tokenizer = AutoTokenizer.from_pretrained(model_id)
    tokenizer.use_default_system_prompt = False
```

Step5: Chat with llama2. Take user prompt, tokenize it and pass to the model. Collect model's response, tokenize it and send to the user.

We define a function `chat_with_llama()` that takes the text prompt as input, converts the text to tokens (`input_ids`) using `tokenizer`. Shifts all `input_ids` to the GPU (`cuda`). It then calls `model.generate()` which takes arguments — `input_ids`, `max_length` etc. A longer `max_length` may require more `cuda` memory so be careful in choosing its value if you are working on free tier Colab.

Then we call this function in a loop to chat with llama2.

```
def chat_with_llama(prompt):
    input_ids = tokenizer.encode(prompt, return_tensors="pt")
    input_ids = input_ids.to('cuda')
    output = model.generate(input_ids, max_length=256, num_beams=4, no_repeat_ngram_size=2)
    response = tokenizer.decode(output[0], skip_special_tokens=True)
    return response

while True:
    prompt = input("You: ")
    response = chat_with_llama(prompt)
    print("Llama:", response)
```

There you go, enjoy you interesting dialogue with llama2. It is very good and accurate just like ChatGPT.

Llama 2

ChatGPT

Python

Chatbots

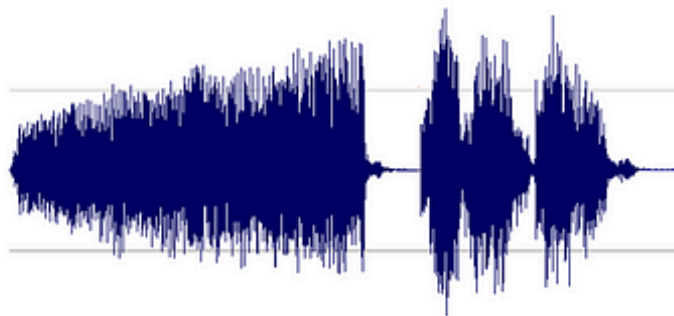
Large Language Models



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Katha Upanishad

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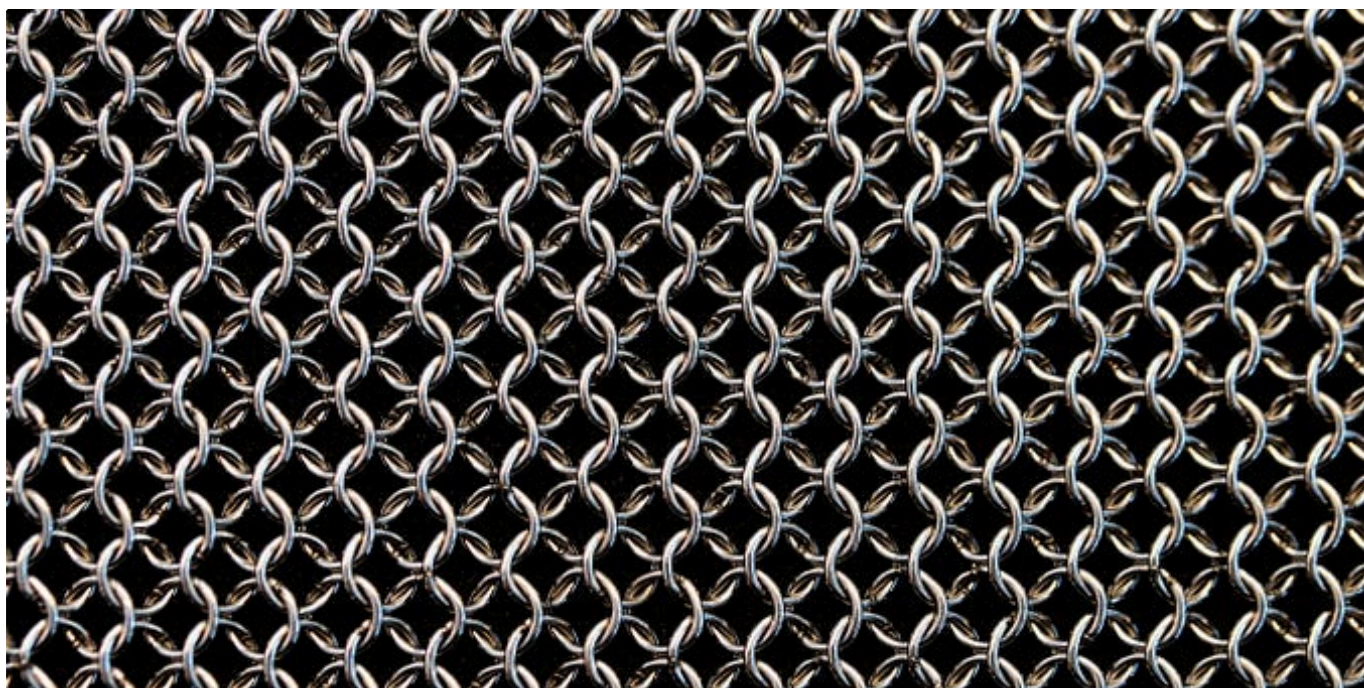


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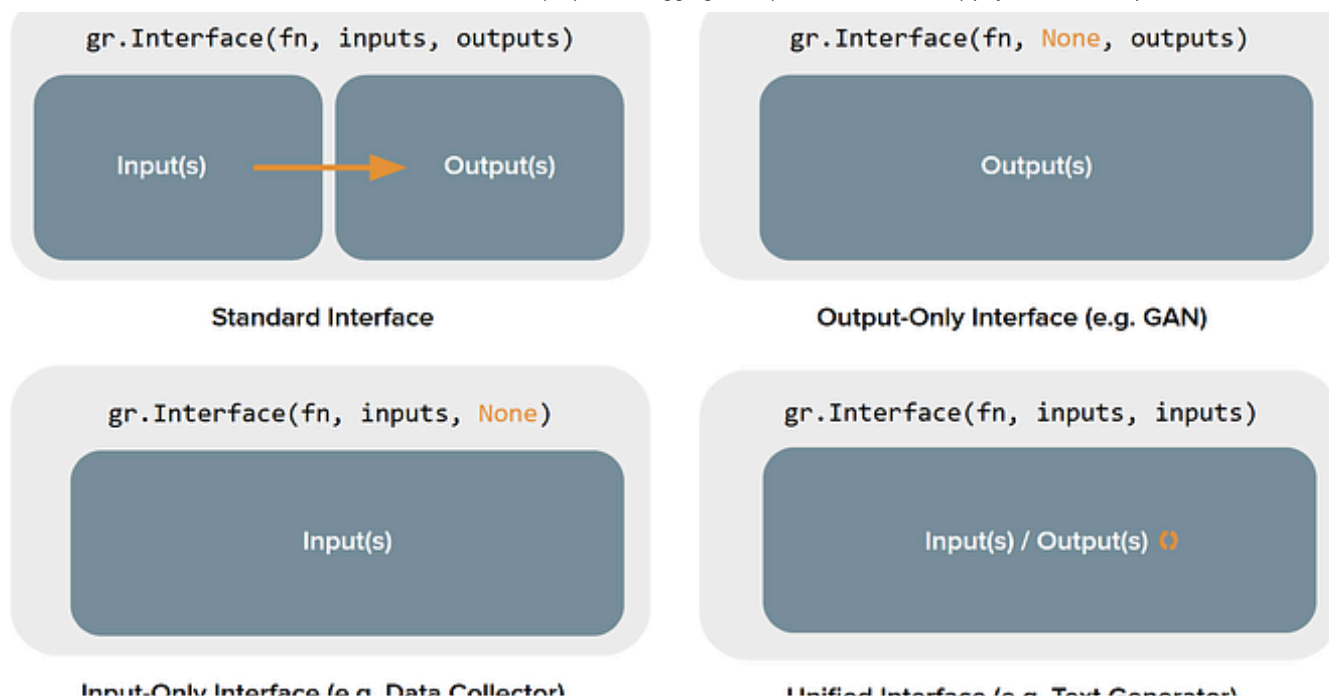
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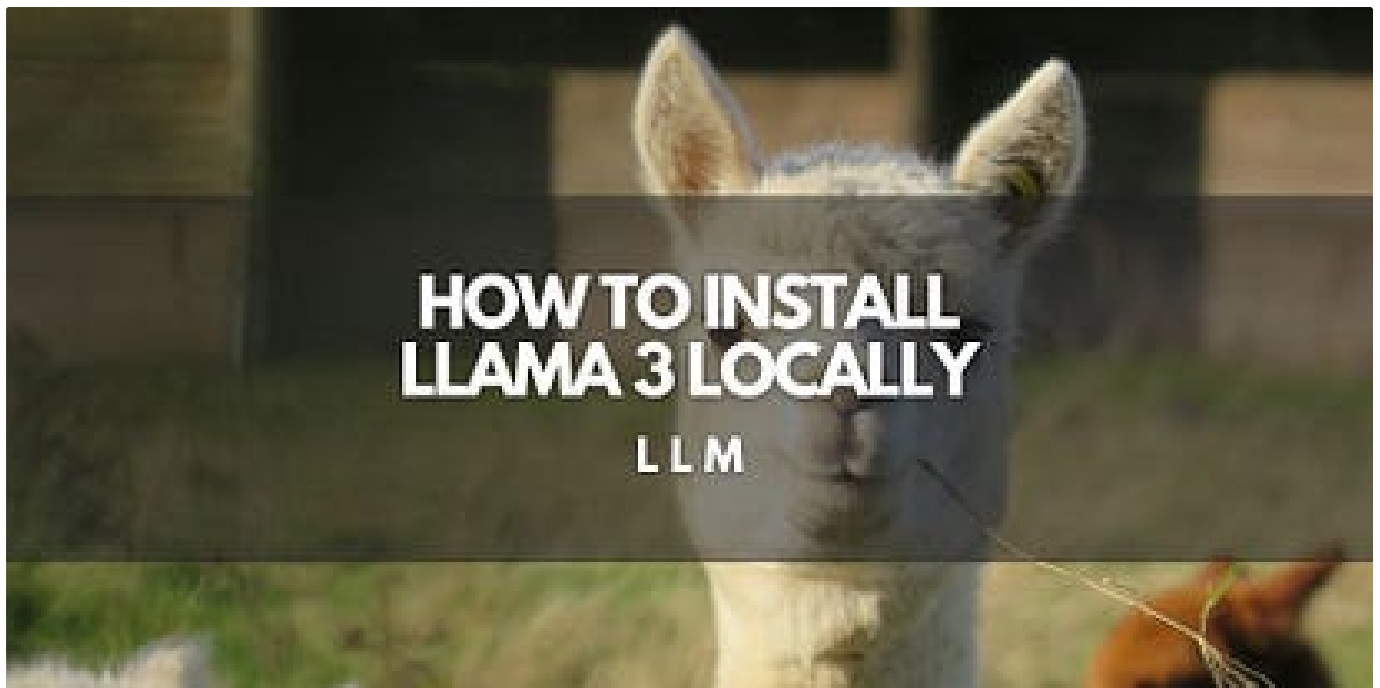
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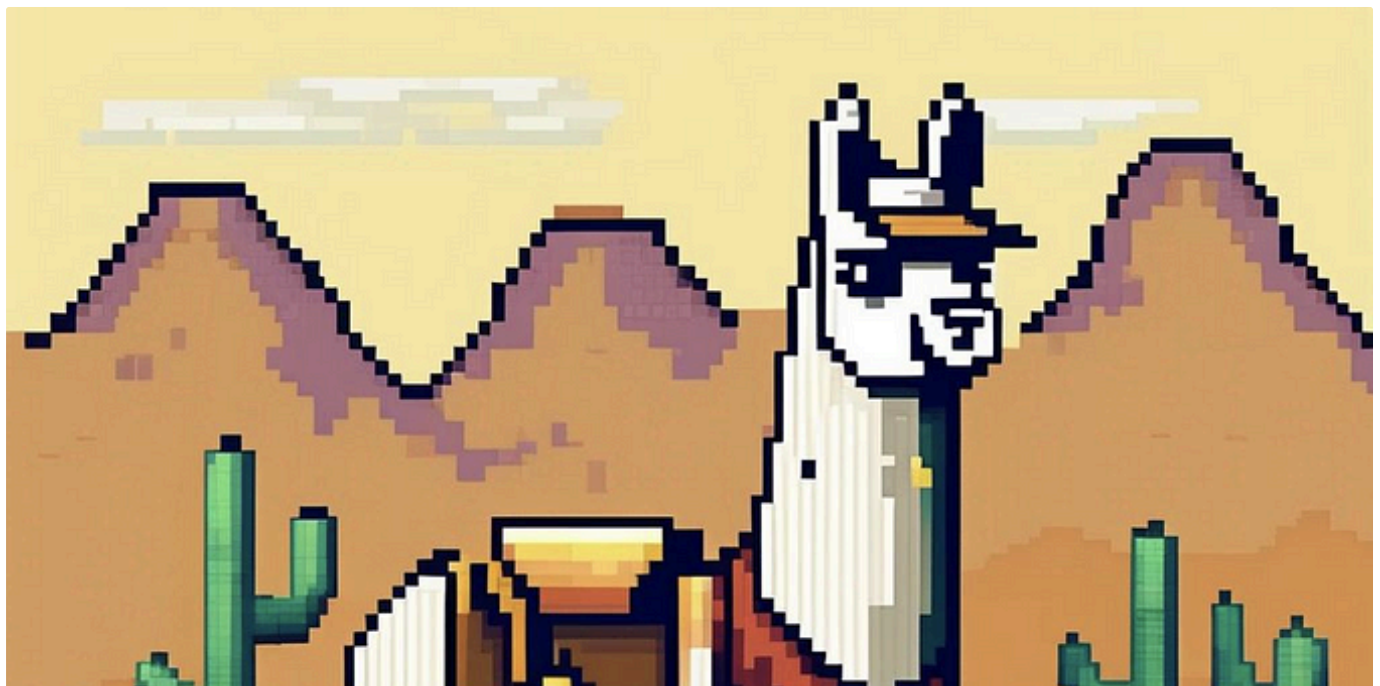


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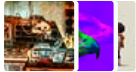
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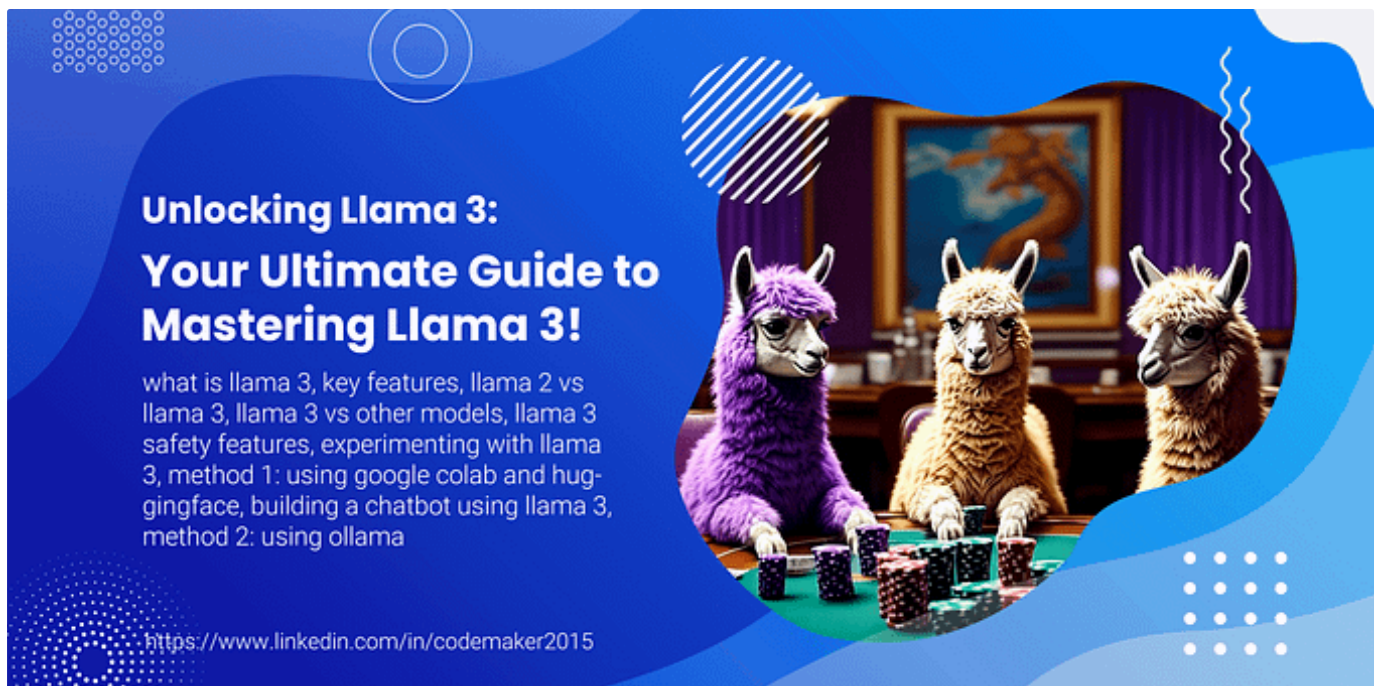
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